

intensive test. We ran the benchmark at 800x600x16 with anti-aliasing off. Content Creation 2001, SiSoft Sandra 2003, PC Mark CPU Test and Adobe Photoshop were also used to evaluate the processors under various application areas. We also used real world tests such as audio and video encoding, and *Quake III Arena*.

As for the motherboards, we used a 256 MB, PC266 266 MHz memory module for testing all DDR boards even though quite a few of them supported DDR333. We had to do this since DDR 333 modules are a rare commodity in India. For RDRAM-based boards, we used a PC 800 memory module except for the ASUS P4T 533-R board that uses a 232 pin PC 800 memory. nVidia's Ti 4200 video chip from ASUS was our choice of graphics card for the systems that had an onboard AGP slot; for those that didn't, we used the onboard video chipset. For testing the boards that support HyperThreading, we used Intel's 3.06 GHz processor, whereas all other boards were tested using the fastest CPU that they supported.

How they fared

With their multi-gigahertz speeds and long pipelines, the Intel Pentium 4 processors are predictably fast. Add in the 512 KB cache and it does not come as a surprise that these pieces of silica shrug off any test thrown at them. They are also supported by good motherboards—Intel chipsets have always been good performers and stable too. Surprisingly chipsets from SiS and VIA are equally good and cheaper by a significant amount.

This does not stop an AMD platform from taking to the soapbox to preach affordability: the processors are fast enough and the motherboards do a decent job of supporting them, all tied in a bundle that is easier on the wallet. Here chipsets from VIA and SiS again shine bright and offer a good set of features and performance.

Gaming and Entertainment

Try playing *Medal of Honor: Allied Assault* on anything below a 700 MHz CPU and watch your system get pounded. You will notice that even if you pair this system with a high horsepower graphics card such as the Ti 4600 or the Radeon 9700 Pro, you still won't experience free-flowing gameplay—the CPU becomes a serious performance bottleneck.

To evaluate gaming performance we used *Quake III Arena* and 3DMark 2001. In the case of entertainment applications—whether you are watching DVD movies, converting your favourite Audio CDs into MP3 or encoding video files—pure CPU power is the call of the day. The video encoding test gives a clear idea of the number crunching ability of a CPU: the ALU

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